Aleksas Mazeliauskas

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Professional experience

2019– Senior research fellow, **Theoretical Physics Department, CERN**

Research on dense and hot QCD matter, Standard Model processes in nuclear environment, future ion experiments at the LHC.

2017–2019 Postdoctoral researcher, **Institute for Theoretical Physics, Heidelberg University**

Member of collaborative research center SFB 1225 *Isolated quantum systems and universality in extreme conditions*.

Work on non-equilibrium QCD physics using quark and gluon kinetic theory, pre-equilibrium kinetic propagator `KøMPøST`, resonance decay code `FastReso`.

2014–2017 Research assistant, **Nuclear Theory Group, Stony Brook University**

PhD dissertation work.

Education

2012–2017 **Stony Brook University**, United States, Doctor of Philosophy in Physics
Max Dresden Prize for outstanding theoretical thesis *Fluctuations in ultra−relativistic heavy ion collisions*, advisor Prof. Derek Teaney, degree awarded on 19/05/2017

2011–2012 **University of Cambridge**, United Kingdom, Master of Mathematics (Part III Maths) graduated with distinction

2008–2011 **University of Cambridge**, United Kingdom, Bachelor of Arts (Mathematical Tripos) 1st class degree

Awards and distinctions

05/05/2022 6-year junior research group leader grant from the Emmy Noether Programme of German Research Foundation (DFG) (accepted)

22/02/2021 6-year junior researcher position from the Portuguese Foundation for Science and Technology (FCT) (declined)

09/11/2019 Nuclear Physics A Young Scientist Award for the best theory talk at Quark Matter 2019

02/05/2017 Max Dresden Prize for outstanding theoretical thesis

25/07/2016 APS FGSA Travel Award for Excellence in Graduate Research

23/05/2013 David Fox award for the outstanding Teaching Assistant

Professional services

Events organised

2022 Co-organiser of the second edition of **The International Conference on Quantum Systems in Extreme Conditions (QSEC2022)**, 14-18 November 2022, Bingen, Germany.

Co-organiser of (virtual) **Second Lithuanian Particle Physics Meeting**, 11 April 2022.

2021 Co-organiser of (virtual) workshop on **Opportunities of OO and pO collisions at the LHC**, 4-10 Feb 2021 and the co-author of the summary report.

Co-organiser of (virtual) **First Lithuanian Particle Physics Meeting**, 9 March 2021.

2019 Convener of discussion group on *Connections between QGP-like observables in small systems* at **International Workshop on QCD Challenges from pp to AA**, 19-23 Aug 2019, Lund, Sweden, and the editor of the summary report.

Convener of discussion group on *Implementations of critical dynamics* at **EMMI Rapid Reaction Task Force: Dynamics of critical fluctuations**, 8-12 April 2019, Darmstadt, Germany, and the editor of the summary report.

Refereeing

Physical Review Letters, Physical Review C and D, Physics Letters B, Nuclear Physics B, Journal of High Energy Physics

Teaching

Heidelberg University

2018-2019 **Co-Head tutor** for Quantum Field Theory and Advanced Quantum Field Theory courses. Responsible for preparing homework and exam problems, typesetting lecture notes and overseeing other tutors (winter and summer semesters).

2018 **Co-organiser** of a master level student lead seminar on QCD matter in heavy ion collisions (summer semester).

2017-2018 **Head tutor** for Quantum Field Theory course. Responsible for preparing homework, exam problems and overseeing nine other tutors (winter semester).

Stony Brook University

2014 **Recitation instructor** for calculus based undergraduate electromagnetism and mechanics courses (2 semesters). Selected student responses:

I think the instructor was more valuable than the course itself.

I would love to see him become a professor one day and attain his career goals.

He would do a fantastic job in teaching the entire course.

2012–2013 **Teaching assistant** for undergraduate mechanics laboratory (2 semesters), received David Fox award for outstanding Teaching Assistant.

National Student Academy (NMA)

2020– Head of Physics section at NMA – additional training school for gifted high-school children in Lithuania. Responsible for the distant learning program and in-person sessions.

2008–2011 Trained a number of young gifted Lithuanian students for physics competitions through distant learning programs and lecturing at summer schools.

Invited lectures

2021 two lectures at **CERN-Fermilab Hadron Collider Physics Summer School** (online) on *High-Density QCD with Proton & Ion Beams*, Geneva, Switzerland

Mentoring

2020 Supervised a visiting PhD student Robin Törnkvist (Lund University)

2019 Informal mentoring of PhD students Oscar Garcia-Montero (Heidelberg University), Giuliano Giacalone (Université Paris Saclay) and Aleksandr Mikheev (Heidelberg University).

2018 Co-supervisor of a bachelor project by Paul Wiesemeyer (Heidelberg University) on radiative photon production in QGP.

Outreach

2022 Organised the display of *Forces of Nature* poster series celebrating women in Physics at CERN Theoretical Physics department.

2021 Conducted a virtual CMS tour and gave a talk about CERN during NMA winter session.

2020 Gave two talks to high-school children in Lithuania about science and work at CERN.

2019 Was interviewed about my work for French science magazine: J.-B. Veyrieras, *Découverte de la première flèche du temps*, Science & Vie, n°1228 (2020)

Prepared physics questions for an on-line science quiz competition *Išmanioji diena* held on 10/10/2019, Lithuania

2017 Prepared two introductory problem sets on elementary particles for high school students at Vilnius University Particle physics outreach group

2008-2013 One of founders and organisers of student-run physics team competition at Vilnius Lyceum, created a number of physics problems for the competition.

2007-2012 Active alumnus of NMA; spoke twice on Lithuanian national TV about education of talented children.

Languages

Lithuanian	native, enthusiastic teacher	**French**	elementary (A2)
English	fluent, higher education in English	**Russian**	basic conversational
German	upper intermediate (B2)	**Chinese**	beginner, passed HSK level II test

Programming

Programming Extensive numerically oriented coding in `C++` and `Python`, previous experience with `Fortran` and Matlab, good working knowledge of Mathematica

General IT Git (source code management), LaTeX (word processing), Inkscape (vector graphics), GNU/Linux (operating system), Gnuplot (graphing), HTML5 (web building)

Invited presentations

22/07/2022 talk at **Predictions for sPHENIX RBRC workshop (remote)** on *Jet and hadron nuclear modification factors*, BNL, United States

19/05/2022 talk at **X Large Hadron Collider Physics Conference (online)** on *Energy loss in small systems*, Taipei, Taiwan

13/05/2022 plenary talk at **Student Conference Mathematics and Natural Sciences: Theory and Application** on *Dalelių fizikos tyrimai CERN: naujų fundamentalių reiškinių ir dalelių paieškos*, Kaunas, Lithuania

25/11/2021 review talk at **III Italian Workshop about high energy physics with heavy ions** on *Theoretical overview on space-time evolution of heavy-ion collisions and QGP temperature*, Padova, Italy

31/08/2021 02/09/2021 two lectures at **CERN-Fermilab Hadron Collider Physics Summer School** (online) on *High-Density QCD with Proton & Ion Beams*, Geneva, Switzerland

30/07/2021 talk at **INT Program: Probing QCD at High Energy and Density with Jets** (online) on *Equilibration and collective effects in QCD kinetic theory*, Seattle, Washington, United States

27/07/2021 review talk at **European Physical Society Conference on High Energy Physics** (online) on *High-energy QCD Matter Theory*, Hamburg, Germany

28/06/2021 plenary talk at **Strong and Electro-Weak Matter Conference** (online) on *Matching glasma to hydro via kinetic theory*, Paris, France

03/03/2021 talk at **145th LHC Comittee Meeting** (online) on *Opportunities of OO and pO collisions at the LHC*, Geneva, Switzerland

15/01/2021 plenary talk at **VI Initial Stages Conference** (online) on *What do we learn from small systems about the physics of heavy ion collisions?*, Rehovot, Israel

24/09/2020 talk at **ALICE Physics Week (online)** on *Partonic rescattering in light nucleus collisions*, Bucharest, Hungary

28/05/2020 talk at **VIII Large Hadron Collider Physics Conference (online)** on *Equilibration of QGP in small systems*, Paris, France

06/12/2019 talk at **XIX Zimányi School Winter Workshop** on *Pre-equilibrium phenomena in Quark Gluon Plasma*, Budapest, Hungary

25/11/2019 talk at **Theoretical Foundations of Relativistic Hydrodynamics Workshop** on *Pre-scaling, hydrodynamic attractors and entropy production in heavy ion collisions*, Banff, Canada

01/11/2019 talk at **New Development of Hydrodynamics and its applications in Heavy-ion Collisions Workshop** on *Hydrodynamic attractors, initial state energy and particle production in relativistic nuclear collisions*, Shanghai, China

13/09/2019 talk at **XLIX International Symposium on Multiparticle Dynamics** on *Equilibration in Quark Gluon Plasma*, Santa Fe, New Mexico, USA

25/06/2019 plenary talk at **V Initial Stages Conference** on *Matching of initial conditions to hydro evolution*, New York, USA

06/04/2019 talk at **XIV Polish Workshop on Relativistic Heavy-Ion Collisions** on *Chemical equilibration in hadronic collisions*, Krakow, Poland

01/08/2018 parallel talk at **XIII Quark Confinement and the Hadron Spectrum Conference**, *2+1D simulations of pre-equilibrium stage with QCD kinetic theory*, Dublin, Ireland

18/05/2018 plenary talk at **XXVII Quark Matter Conference** on *Initial conditions for nuclear collisions: theory overview*, Venice, Italy

02/10/2017 talk at **XII High-pT Physics Workshop** on *Initial conditions for heavy ion collisions* , Bergen, Norway

15/06/2017 talk at **XII Workshop on Particle Correlations and Femtoscopy** on *Kinetic theory equilibration for realistic heavy ion initial conditions*, Amsterdam, The Netherlands

06/03/2015 talk at **Collectivity in Small Colliding Systems with High Multiplicity Workshop** on *Principal Component Analysis and Subleading Flow*, Brookhaven National Lab, New York, United States

Contributed presentations

06/04/2022 **XXIX Quark Matter Conference**, Krakow, Poland

07/10/2021 **XLIV National Lithuanian Physics Conference**, Vilnius, Lithuania

09/03/2021 **First Lithuanian Particle Physics Meeting** (online), Vilnius, Lithuania

03/06/2020 **X Hard Probes (online)**, Austin, USA

06/11/2019 **XXVIII Quark Matter Conference**, Wuhan, China, BEST THEORY TALK

22/09/2019 **Quantum Systems in Extreme Conditions**, Heidelberg, Germany
10/06/2019 **XVIII Strangness in Quark Matter**, Bari, Italy
14/01/2019 **XLVII Workshop on Gross Properties of Nuclei and Nuclear Excitations**, Hirschegg, Austria
23/05/2019 **Origins of Correlations in High Energy Collisions**, INT, Seattle, Washington, United States
09/04/2019 **EMMI RRTF: Dynamics of critical fluctuations**, Darmstadt, Germany
02/10/2018 **IX Hard Probes conference**, Aix-Les-Bains, France
27/08/2018 **Probing the Quark-Gluon Plasma with collective phenomena and heavy quarks**, MIAPP, Munich, Germany
19/09/2017 **IV Initial Stages**, Krakow, Poland
08/08/2017 **Critical Point and Onset of Deconfinement**, Stony Brook, New York, United States
07/02/2017 **XXVI Quark Matter**, Chicago, Illinois, United States
19/09/2016 **VII Hot Quarks**, South Padre Island, Texas, United States
25/05/2016 **III Initial Stages**, Lisbon, Portugal
22/07/2015 **Correlations and Fluctuations in p+A and A+A Collisions**, INT, Seattle, Washington, United States
27/03/2015 **Ohio-Region APS meeting**, Kent, Ohio, United States

List of publications

2022

1. M. Attems, J. Brewer, G. M. Innocenti, A. Mazeliauskas, S. Park, W. van der Schee and U. A. Wiedemann, *The medium-modified $g \to c\bar{c}$ splitting function in the BDMPS-Z formalism*, arXiv:2203.11241
2. A. N. Mikheev, A. Mazeliauskas and J. Berges, *Stability analysis of non-thermal fixed points in longitudinally expanding kinetic theory*, **Phys. Rev. D** 105, 116025 (2022), arXiv:2203.02299

2021

3. W. Florkowski, A. Kumar, A. Mazeliauskas, R. Ryblewski, *Effect of thermal shear on longitudinal spin polarization in a thermal model*, **Phys. Rev. C** 105, 064901 (2022), arXiv:2112.02799
4. J. Brewer, A. Huss, A. Mazeliauskas, W. van der Schee, *Ratios of jet and hadron spectra at LHC energies: measuring high-p_T suppression without a pp reference*, **Phys.Rev.D** 105, 074040 (2022), arXiv:2108.13434
5. C. Duhr, A. Huss, A. Mazeliauskas, R. Szafron, *Analysis of Bayesian estimates for missing higher orders in perturbative calculations*, **JHEP** 09 (2021) 122, arXiv:2106.04585, https://doi.org/10.1007/JHEP09(2021)122
6. A. Andronic, P. Braun-Munzinger, M.K. Köhler, A. Mazeliauskas, K. Redlich, J. Stachel, V. Vislavicius, *The multiple-charm hierarchy in the statistical hadronization model*, **JHEP** 07 (2021) 035, arXiv:2104.12754, https://doi.org/10.1007/JHEP07(2021)035
7. A. Kurkela, A. Mazeliauskas, R. Törnkvist, *Collective flow in single-hit QCD kinetic theory*, **JHEP** 11 (2021) 216, arXiv:2104.08179, https://doi.org/10.1007/JHEP11(2021)216
8. J. Brewer, A. Mazeliauskas, W. van der Schee, *Opportunities of OO and pO collisions at the LHC*, arXiv:2103.01939

2020

9. A. Huss, A. Kurkela, A. Mazeliauskas, R. Paatelainen, W. van der Schee, U. Wiedemann, *Predicting parton energy loss in small collision systems*, **Phys.Rev.C** 103 054903 (2021) arXiv:2007.13758, https://doi.org/10.1103/PhysRevC.103.054903

10. A. Huss, A. Kurkela, A. Mazeliauskas, R. Paatelainen, W. van der Schee, U. Wiedemann, *Discovering partonic rescattering in light nucleus collisions*, **Phys.Rev.Lett.** 126 192301 (2021), arXiv:2007.13754, https://doi.org/10.1103/PhysRevLett.126.192301
11. J. Berges, M.P. Heller, A. Mazeliauskas, R. Venugopalan, *QCD thermalization: Ab initio approaches and interdisciplinary connections*, **Rev.Mod.Phys.** 93 (2021) 3 035003 arXiv:2005.12299, https://doi.org/10.1103/RevModPhys.93.035003
12. EDITOR FOR J. Adolfsson et al., *QCD Challenges from pp to A-A Collisions*, community report, **Eur.Phys.J.A 56, 11 (2020)**, arXiv:2003.10997
13. EDITOR FOR M. Bluhm et al., *Dynamics of critical fluctuations: Theory – phenomenology – heavy-ion collisions*, community report, **Nucl. Phys. A** 1003, 122016 (2002), arXiv:2001.08831

2019

14. O. Garcia-Montero, N. Löher, A. Mazeliauskas, J. Berges, and K. Reygers, *Probing the evolution of heavy-ion collisions using direct photon interferometry*, **Phys. Rev. C** 102, 024915 (2020), arXiv:1909:12246, https://doi.org/10.1103/PhysRevC.102.024915
15. D. Devetak, A. Dubla, S. Floerchinger, E. Grossi, A. Mazeliauskas, S. Masciocchi, and I. Selyuzhenkov, *Global fluid fits to identified particle transverse momentum spectra from heavy-ion collisions at the Large Hadron Collider*, **JHEP** 2006 (2020) 44 arXiv:1909:10485, https://doi.org/10.1007/JHEP06(2020)044
16. P. Hanus, A. Mazeliauskas, and K. Reygers, *Entropy production in pp and Pb-Pb collisions at energies available at the CERN Large Hadron Collider*, **Phys. Rev. C** 100, 064903 (2019), arXiv:1908.02792, https://doi.org/10.1103/PhysRevC.100.064903
17. G. Giacalone, A. Mazeliauskas and S. Schlichting, *Hydrodynamic Attractors, Initial State Energy, and Particle Production in Relativistic Nuclear Collisions*, **Phys. Rev. Lett.** 123, 262301 (2019), arXiv:1908.02866,https://doi.org/10.1103/PhysRevLett.123.262301
18. A. Mazeliauskas and V. Vislavicius, *Temperature and fluid velocity on the freeze-out surface from π, K, p spectra in pp, p–Pb and Pb–Pb collisions*, **Phys. Rev. C** 101, 014910 (2020), arXiv:1907.11059, https://doi.org/10.1103/PhysRevC.101.014910
19. W. Florkowski, A. Kumar, R. Ryblewski, and A. Mazeliauskas, *Longitudinal spin polarization in a thermal model*, **Phys. Rev. C 100**, 054907 (2019), arXiv:1904.00002, https://doi.org/10.1103/PhysRevC.100.054907

2018

20. A. Kurkela, A. Mazeliauskas, *Chemical equilibration in weakly coupled QCD*, **Phys. Rev. D** 99, 054018 (2019), EDITORS' SUGGESTION, arXiv:1811.03068, https://doi.org/10.1103/PhysRevD.99.054018
21. A. Kurkela, A. Mazeliauskas, *Chemical equilibration in hadronic collisions*, **Phys. Rev. Lett.** 122, 142301 (2019), arXiv:1811.03040, https://doi.org/10.1103/PhysRevLett.122.142301
22. A. Mazeliauskas, J. Berges, *Prescaling and far-from-equilibrium hydrodynamics in the quark-gluon plasma*, **Phys. Rev. Lett.** 122, 122301 (2019), arXiv:1810.10554, https://doi.org/10.1103/PhysRevLett.122.122301
23. A. Mazeliauskas, S. Floerchinger, E. Grossi, D. Teaney, *Fast resonance decays in nuclear collisions*, **Eur. Phys. J. C** (2019), arXiv:1809.11049, https://doi.org/10.1140/epjc/s10052-019-6791-7

24. A. Kurkela, A. Mazeliauskas, J.-F. Paquet, S. Schlichting and D. Teaney, *Matching the non-equilibrium initial stage of heavy ion collisions to hydrodynamics with QCD kinetic theory*, **Phys. Rev. Lett.** 122, 122302 (2019), EDITORS' SUGGESTION, arXiv:1805.01604, https://doi.org/10.1103/PhysRevLett.122.122302
25. A. Kurkela, A. Mazeliauskas, J.-F. Paquet, S. Schlichting and D. Teaney, *Effective kinetic description of event-by-event pre-equilibrium dynamics in high-energy heavy-ion collisions*, **Phys. Rev. C,99**, 034910 (2019), arXiv:1805.00961, https://doi.org/10.1103/PhysRevC.99.034910

pre-2018

26. Y. Akamatsu, A. Mazeliauskas and D. Teaney, *Bulk viscosity from hydrodynamic fluctuations with relativistic hydro-kinetic theory*, **Phys. Rev. C** 97, 024902 (2018), arXiv:1708.05657, https://doi.org/10.1103/PhysRevC.97.024902,
27. Y. Akamatsu, A. Mazeliauskas and D. Teaney, *A kinetic regime of hydrodynamic fluctuations and long time tails for a Bjorken expansion*, **Phys. Rev. C**,95, 014909 (2017), arXiv:1606.07742 https://doi.org/10.1103/PhysRevC.95.014909
28. L. Keegan, A. Kurkela, A. Mazeliauskas and D. Teaney, *Initial condition for hydrodynamics from weakly coupled pre-equilibrium evolution*, **JHEP** 1608 (2016) 171, arXiv:1605.04287, https://doi.org/10.1007/JHEP08(2016)171
29. A. Mazeliauskas and D. Teaney, *Fluctuations of harmonic and radial flow in heavy ion collisions with principal components*, **Phys. Rev. C** 93, 024913 (2016), arXiv:1509.07492, https://doi.org/10.1103/PhysRevC.93.024913
30. A. Mazeliauskas and D. Teaney, *Subleading harmonic flows in hydrodynamic simulations of heavy ion collisions*, **Phys. Rev. C** 91, 044902 (2015), arXiv:1501.03138, https://doi.org/10.1103/PhysRevC.91.044902

Selected Conference proceedings

1. A. Mazeliauskas, **Quark Matter 2018** *Initial conditions for nuclear collisions: theory overview*, Nucl. Phys. A 982 (2019), 134-141, arXiv:1807.05586